

The synaptic Package: User Manual

synaptic project

Open-source LaTeX community

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Abstract: `synaptic` is a modular LaTeX3 design system for journal articles, books, lecture material, and compact notes. This manual describes its requirements, safe configuration model, metadata API, theorem system, and mode-specific features.

Keywords: `synaptic`, LaTeX3, LuaLaTeX, academic typesetting

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1 Getting started

The package requires LuaLaTeX and a KOMA-Script class. The repository ships generated package files, so a local checkout can be used without installation:

```
export TEXINPUTS=/path/to/synaptic/tex/latex/synaptic//:$TEXINPUTS
lualatex document.tex
```

A minimal article is:

```
\documentclass{scrartcl}
\usepackage[mode=journal]{synaptic}
\SynapticTitle{A Short Article}
\SynapticAuthor{Ada Author}
\begin{abstract}Summary.\end{abstract}
\begin{document}
\SynapticMakeTitle
\section{Introduction}
Text.
\end{document}
```

Use `scrbook` for book mode. The selected class controls the paper size; synaptic does not silently force A4.

2 Configuration

Key	Default	Values and purpose
<code>mode</code>	<code>journal</code>	<code>journal</code> , <code>book</code> , <code>lecture</code> , or <code>notes</code>
<code>theme</code>	<code>ocean</code>	<code>ocean</code> , <code>graphite</code> , <code>forest</code> , <code>midnight</code> , or <code>paper</code>
<code>lang</code>	<code>en</code>	<code>en</code> or <code>zh</code> ; selects labels and CJK support
<code>fontset</code>	<code>auto</code>	<code>auto</code> , <code>xcharter</code> , <code>libertinus</code> , <code>stix2</code> , or <code>lm</code>
<code>toc</code>	<code>false</code>	whether the title workflow prints a TOC
<code>toc-layout</code>	<code>top</code>	<code>top</code> , <code>inline</code> , or <code>page</code>
<code>title-layout</code>	<code>auto</code>	<code>inline</code> , <code>compact</code> , <code>page</code> , <code>cover</code> , or <code>mode default</code>
<code>numbering</code>	<code>auto</code>	<code>section</code> , <code>chapter</code> , <code>continuous</code> , <code>none</code> , or <code>mode default</code>
<code>box-numbering</code>	<code>shared</code>	<code>shared</code> , <code>separate</code> , or <code>none</code>
<code>color-mode</code>	<code>screen</code>	<code>screen</code> , <code>print</code> , or <code>mono</code>
<code>density</code>	<code>balanced</code>	<code>compact</code> , <code>balanced</code> , or <code>airy</code>
<code>measure</code>	<code>auto</code>	<code>narrow</code> , <code>standard</code> , <code>wide</code> , or <code>mode default</code>
<code>binding-offset</code>	<code>0pt</code>	additional book binding correction
<code>extra-math</code>	<code>false</code>	convenience Unicode math alphabets
<code>fine-breaking</code>	<code>true</code>	moderate emergency stretch and tolerance

Only `theme`, `toc`, and `toc-layout` are safe runtime settings:

`\SynapticSetup{theme=paper,toc=false,toc-layout=page}`

Mode, language, font, and load-time typography options are frozen after package loading. Attempts to change them issue a warning instead of producing a half-reconfigured document.

2.1 Modes

Mode	Class	Purpose
journal	scrartcl	Inline article title, metadata, running heads, statements
book	scrbook	Book title sequence, parts, chapters, duplex navigation
lecture	scrartcl/scrreprt	Course masthead and semantic teaching system
notes	scrartcl	Compact masthead and lightweight knowledge cards

2.2 Fonts

The `auto` setting chooses the first *complete* installed bundle: XCharter, Libertinus, STIX Two, then Latin Modern. An explicit choice reports a clear error if a required companion font is absent. Chinese documents prefer Noto Serif/Sans CJK SC and fall back to the Fandol fonts distributed with TeX Live.

3 Title and PDF metadata

All four modes share the metadata model and render it differently:

Command	Meaning
<code>\SynapticTitle</code>	Required printed title and PDF title
<code>\SynapticSubtitle</code>	Optional display subtitle
<code>\SynapticShortTitle</code>	Running-head title
<code>\SynapticDate</code>	Explicit date
<code>\SynapticAuthor</code>	Repeatable author and optional affiliation mark
<code>\SynapticAffiliation</code>	Repeatable affiliation
<code>\SynapticEmail</code>	Repeatable clickable contact address
<code>\SynapticSubject</code>	PDF subject
<code>\SynapticKeywords</code>	Printed and PDF keywords
<code>\SynapticHeader</code>	Submission header override
<code>\SynapticSubHeader</code>	Secondary header
<code>\SynapticPreprint</code>	Preprint identifier
<code>\SynapticArxiv</code>	Linked arXiv identifier
<code>\SynapticDedicated</code>	Dedication
<code>\SynapticMakeTitle</code>	Validate and render all title data

Journal mode accepts publication information, correspondence, and author notes; book mode accepts half-title, publisher, edition, ISBN, and copyright fields; lecture mode accepts course, code, lecture number, instructor, and semester. Use

`\SynapticStatement{heading}{body}` for journal funding, data, ethics, or conflict statements; the starred form omits its table-of-contents entry. Standard `\title`, `\author`, and `\date` are imported when the Synaptic field is empty. PDF metadata receives one Unicode-safe conversion.

4 Theorems and mathematics

The built-in `theorem`, `lemma`, `proposition`, and `definition` environments share one counter. Books use chapter scope by default; shorter modes use sections. Definitions use upright body text; proposition-style statements use italics in English and upright CJK text.

Theorem 4.1 (Gaussian integral). *The integral satisfies*

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

Proof. Square the integral, change to polar coordinates, and evaluate the radial integral. ■

New theorem types accept a style:

`\SynapticNewTheorem[remark]{observation}{Observation}`

The available styles are `plain`, `definition`, and `remark`.

Observation 4.2. With `extra-math=true`, commands such as $\mathbb{R}$, $\mathcal{F}$, and $\mathfrak{g}$ are available.

5 Themes and lecture boxes

Themes may be switched with `\SynapticUseTheme{name}`. A custom theme is defined and activated using `xcolor` expressions:

`\SynapticDefineTheme{brand}{blue!70!black}{gray!80!black}[orange]`

Lecture mode adds the collision-resistant environments `synexample`, `synremark`, `synwarning`, `synexercise`, and `synbox`. Examples and exercises share the theorem counter by default; remarks and warnings are unnumbered. The first optional argument is a title and the second is a label. The generic v2.0 aliases are created only when their names are free. Course units may also use `synlearninggoals` and `synlecturesummary`.

6 Book workflow and acknowledgments

Use `\SynapticFrontMatter`, `\SynapticMainMatter`, and `\SynapticBackMatter` around the corresponding parts of a book. `\SynapticBookContents`, chapter subtitles, epigraphs, and list/appendix helpers complete the long-form workflow. Notes mode provides `synkeyidea`, `synquestion`, `syntodo`, and `synsummary`; class-level two-column layouts are supported. The `\SynapticAcknowledgments` command creates a chapter in book mode and a section in other modes. Its starred form suppresses the TOC entry.

7 Development commands

```
l3build unpack    # regenerate .sty and .def files from synaptic.dtx
l3build check     # run the LuaLaTeX regression suite
l3build doc       # build the documented source and manuals
l3build ctan      # create release archives
```

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